

Introduction
Basic Mech. Engg.

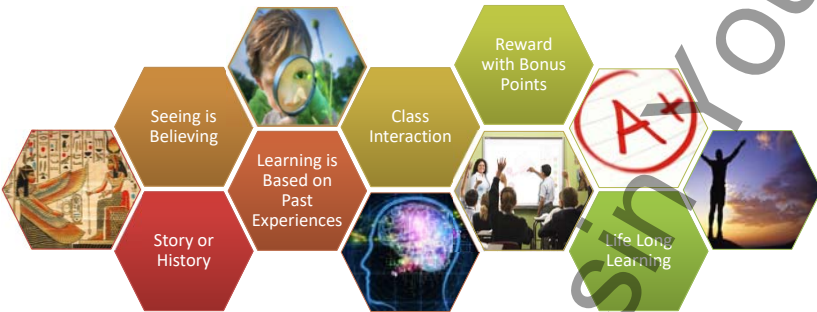
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TEACHING METHODOLOGY



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SLIDE THEME

Questions, Disadvantages &Warnings

Answers & Advantages

Definitions & Facts

[1] - References

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Basic Mechanical Engineering

IDEA

What does "Engineering" Mean?
Application of scientific, economic, social, and practical knowledge in order to invent, design, build, maintain, research, and improve structures, machines, devices, systems, materials and processes.

What is Mechanical Engineering?
Discipline that applies the principles of engineering, physics and materials science for the design, analysis, manufacturing, and maintenance of mechanical systems

What is meant by Mechanical Systems?
A mechanical system manages power to accomplish a task that involves forces and movement.

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ENGINEER

What does an Engineer do? → ? Cleverness
Applies scientific knowledge and ingenuity to develop solutions for technical, societal and commercial problems

What are you doing here?
Learning scientific knowledge and its applications

YOU ALL ARE PROBLEM SOLVERS



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APPLICATIONS



The Astounding Athletic Power of Quad copters

[1] - Raffaello D'Andrea: The astounding athletic power of quadcopters, TEDGlobal (JUN 2013). URL: <http://on.ted.com/Quadcopter>

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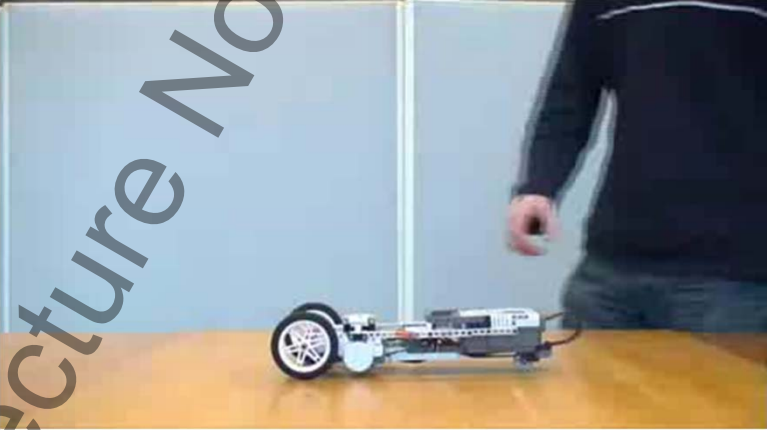
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APPLICATIONS



Engineering Mechanics is the study of forces and movements [1]

[1] - Youtube - January 1, 2015 (www.youtube.com/watch?v=RFfIdyojDmk)

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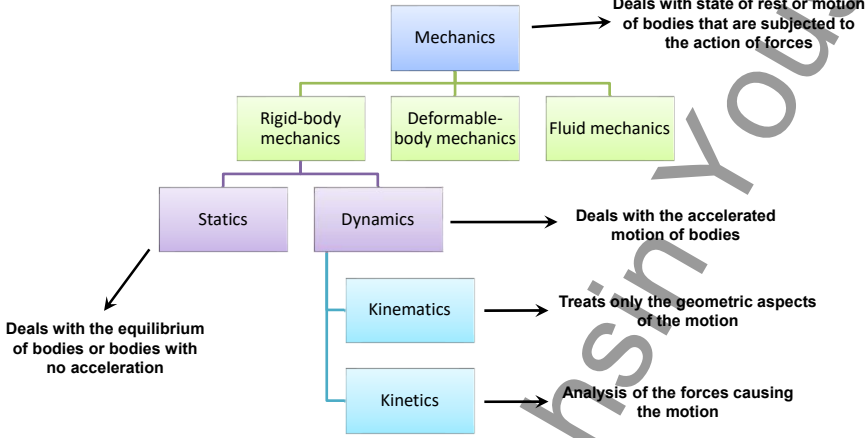
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ENGINEERING MECHANICS



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graph TD
    Mechanics[Mechanics] --> RBM[Rigid-body mechanics]
    Mechanics --> DBM[Deformable-body mechanics]
    Mechanics --> FM[Fluid mechanics]
    RBM --> Statics[Statics]
    RBM --> Dynamics[Dynamics]
    Statics --> SDesc[Deals with the equilibrium of bodies or bodies with no acceleration]
    Dynamics --> Kinematics[Kinematics]
    Dynamics --> Kinetics[Kinetics]
    Kinematics --> KDesc[Treats only the geometric aspects of the motion]
    Kinetics --> KDesc2[Analysis of the forces causing the motion]
    Dynamics --> DDesc[Deals with the accelerated motion of bodies]
  
```

Deals with state of rest or motion of bodies that are subjected to the action of forces

Deals with the equilibrium of bodies or bodies with no acceleration

Deals with the accelerated motion of bodies

Treats only the geometric aspects of the motion

Analysis of the forces causing the motion

Essential for the design and analysis of many types of structural members, mechanical components, or electrical devices encountered in engineering

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OBJECTIVE

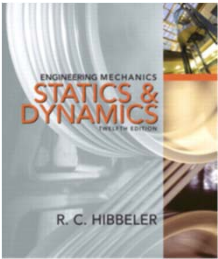
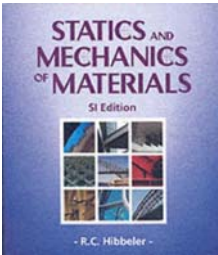
To build the fundamental concepts of engineering mechanics and to clearly present their application in the analysis of planner motion of mechanical components and structures.

COURSE BOOK

- ✓ *Engineering Mechanics: Statics and Dynamic*, by R.C Hibbeler, Pearson Prentice Hall, 12th Edition, 2010.

REFERENCE BOOKS

- ✓ *Statics and Mechanics of Materials*, by R.C Hibbeler, Pearson Prentice Hall, SI Edition, 2004.
- ✓ *Engineering Mechanics: Statics and Dynamics*, by J. L. Meriam, and L. G. Kraige, John & Wiley Sons , 7th Edition,, 2012

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COURSE OUTLINE

Week Number	Book Topics Numbers	Topic Names
1	1.1 to 1.3	Introduction to Vector Mechanics + A1
2	2.1 to 2.7	Force Systems
3	3.1 to 3.2	Free Body Diagram + Quiz I
4	3.5 to 3.10	Equilibrium of a Rigid Body
5	5.1 to 5.2	Friction
6	Midterm Examination I	

Appropriate changes would be made if required.

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COURSE OUTLINE

Week Number	Book Topics Numbers	Topic Names
7	16.1 to 16.2	Planar Kinematics: Absolute Motion Analysis
8	16.3 to 16.4	Planar Kinematics: Absolute Motion Analysis
9	16.5	Planar Kinematics : Relative Motion Analysis
10	16.6	Planar Kinematics : Relative Motion Analysis+ Quiz II
11	16.7	Planar Kinematics : Relative Motion Analysis
12	Midterm Examination II	

Appropriate changes would be made if required.

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COURSE OUTLINE

Week Number	Book Topics Numbers	Topic Names
13	16.1 to 16.4	Mechanics of Materials: Stress and Strain
14	16.5, 16.7	Mechanics of Materials: Torsion & Bending
15	17	Mechanical Properties of Material + Quiz III
16	18	Structure Analysis: Frames and Machines
		Final Examination

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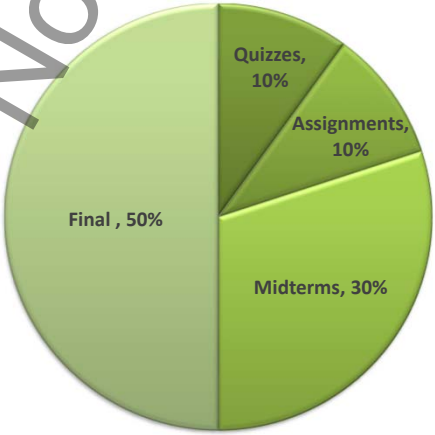
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GRADING POLICY



Category	Percentage
Final	50%
Midterms	30%
Quizzes	10%
Assignments	10%
Bonus	5% Absolute (Max)

Bonus, 5% Absolute (Max)

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A+ OR F

How to succeed in this course?

- ✓ Arrive on time
- ✓ Participate in all classes
- ✓ Pay attention, take notes and ask questions if needed
- ✓ Identify and understand key issues
- ✓ Review material (notes and textbook) before and after class
- ✓ Avoid Plagiarism
- ✓ Practice, Practice and Practice



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Take-Home Message

THOUGHT

You are a Problem Solver



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